Sathira Silva

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Interests

3D Perception | Vision-Language Models | Self-supervised Learning

Education

University of Peradeniya, Sri Lanka **Nov. 2018 – Dec. 2023**
B.Sc. (Hons) in Computer Engineering *GPA: 3.60 / 4.00*
[Courses, Thesis] *Class: 2nd Upper*

De Mazenod College **Jan. 2003 – August 2016**
G.C.E. Advanced Level *Maths - A, Physics - A, Chemistry - B*

Experience

Research Engineer **May 2025 – Current**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Lead the developmentally inspired VLM training research: BabyMind where we tried to outperform existing work CVCL via infant-inspired inductive biases.
- Contributed to a domain generalizable visual recognition research direction using baby videos.
- Contributing to developing a monotone-order based compressed structural encoding for style-normalized pathology-faithful CXR reconstruction towards domain adaptability.
- Contributing to developing an efficient caching method for generating surgical videos.

Research Assistant **June 2024 – April 2025**
Computer Vision Department — Principal Investigator: *Dr. Muhammad Haris Khan* *MBZUAI*

- Contributed to developing a SOTA solution (DPA) to unsupervised fine-tuning of VLMs (CLIP) using *novel dual prototypes*.
- Fabricated a *novel multi-crop-based self-training pipeline* for fine-grained unsupervised adaptation of CLIP using fine-grained interactions between the two modalities, outperforming the SOTA.
- Worked on a *novel method for uncovering CLIP's hidden fine-grained knowledge* for unsupervised adaptation of CLIP on fine-grained datasets.

Computer Vision Research Engineering Intern **Dec. 2022 – Mar. 2023**
Autonomous Vehicle R&D Division *Vega Innovations*

- Contributed to the integration of a transformer architecture called NEAT into an autonomous vehicle system, by reviewing the paper and understanding its internals.
- Developed real-time computer vision solutions for autonomous vehicles on high performance GPU inference embedded systems (Nvidia DRIVE PX2 / Jetson TX2).

Teaching Assistant: Programming Methodology (CO222) **May 2021 – Sep. 2021**
Department of Computer Engineering *University of Peradeniya*

- Supervised weekly 2hr long online lab sessions.
- Created questions for online quizzes based on the C programming Language.
- One-on-one sessions with students to tutor them on the C programming language concepts.

Selected Achievements

ACES Coders v7.0 | An inter-university algorithmic programming competition organized by of UoP **2020**
Rank - 14 / 100+
Team Name: bitLasagna

IEEEXtreme 14.0 | 24-hour global algorithmic programming competition **2020**
Country Rank - 2, Global Rank - 68 / 7000+
Team Name: InterGreat

ICDS Mini Hackathon | An inter-university Data Science Hackathon **2021**
Rank - 5 / 100+
Team Name: bitLasagna

IEEEXtreme 16.0 | 24-hour global algorithmic programming competition **2022**
Country Rank - 27, Global Rank - 427 / 6373
Team Name: bitLasagna

ACES Coders v9.0 | An inter-university algorithmic programming competition organized by of UoP **2022**
Rank - 2 / 100+
Team Name: bitLasagna

Publications

S2TPVFormer: A Spatiotemporal Approach to Tri-Perspective Representation for 3D Semantic Occupancy Prediction arXiv poster **Dec. 2024**
[*Best paper award*] In AAAI 2025 Machine Learning for Autonomous Driving (ML4AD) Workshop
Authors: Sathira Silva, Savindu Wannigama*, Gihan Jayatilake, Muhammad Haris Khan, Roshan Ragel*

microCLIP: Unsupervised CLIP Adaptation via Coarse-Fine Token Fusion for Fine-Grained Image Classification arXiv **Dec. 2025**
Accepted at ACL (Findings) 2026
Authors: Sathira Silva, Eman Ali, Chetan Arora, Muhammad Haris Khan

Objects Before Words: Object-First Inductive Biases for Grounding Language in Child-View Video **Feb. 2026**
Accepted at CogSci 2026
Authors: Sathira Silva, A. K. Gebreselasie, M. Umer Sheikh, Kartik Kuckreja, Daniel Harari, M. H. Khan

Towards Fine-Grained Adaptation of CLIP via a Self-Trained Alignment Score arXiv **July 2025**
[Oral] In 2026 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
Authors: Eman Ali, Sathira Silva, Chetan Arora, Muhammad Haris Khan

DPA: Dual Prototypes Alignment for Unsupervised Adaptation of Vision-Language Models arXiv poster **Oct. 2024**
In 2024 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
Authors: Eman Ali, Sathira Silva, Muhammad Haris Khan

Learning Visual Referents from Noisy Child-view Videos **Apr. 2026**
Under Review in ACM Multimedia 2026
Authors: Abrham Kahsay Gebreselasie, Sathira Silva Michael Sidorov, Daniel Harari, Muhammad H. Khan

Ongoing Projects

Monotone-Order Strctural Sketch (MOSS) for Domain-Generalizable Chest X-ray Diagnosis **Current**
Index Terms: Domain Generalization, Medical Imaging, Image Harmonization *Research*
Technologies: PyTorch, Distributed Training

- Developing a plug-and-play chest X-ray harmonization framework that develops an efficient structural encoding with semantic decoder adapters to preserve pathology-relevant evidence while reducing scanner- and site-specific style variation.

MedCache: Caching for Efficient Medical World Model Inference **Current**
Index Terms: Medical World Models, Video Generation, Diffusion Caching *Research*
Technologies: PyTorch, Distributed Inference, VBench / Med-VBench

- Developing a caching-based inference framework for medical world models to reduce video generation latency while preserving clinical visual fidelity and temporal consistency.

Technical Skills

Languages: C/C++, Python, Java, HTML/CSS, JavaScript, SQL

Developer Tools: Visual Studio, VS Code, Eclipse, Jupyter Notebook, Android Studio

Technologies/Frameworks: OpernMMLab, PyTorch, TensorFlow, Bash Scripting, GitHub, OpenCV, TensorFlow, ReactJS, NodeJS, Jekyll

Certifications

Natural Language Processing (hons) **Jan. 2022**
HSE University *Coursera*

Algorithms on Graphs **July 2020**
University of California San Diego *Coursera*

Data Structures **June 2020**
University of California San Diego *Coursera*

Convolutional Neural Networks **Feb. 2020**
DeepLearning.AI *Coursera*

Neural Networks and Deep Learning **Jan. 2020**
DeepLearning.AI *Coursera*

Relevant Coursework

Data Structures & Algorithms	Operating Systems
Software Methodology	Computer Architecture
Image Processing	Programming Methodology
Artificial Intelligence	Discrete Mathematics
Networking and Web Application Design	Probability and Statistics

References

Dr. Muhammad Haris Khan

Assistant Professor,
Department of Computer Vision,
Mohammed bin Zayed University of AI,
United Arab Emirates

Prof. Roshan G. Ragel

Head of Department,
Department of Computer Engineering,
Faculty of Engineering,
University of Peradeniya, Sri Lanka